Image-Based MF-ATI and PSMF-ATI with Array Radar GMTI Systems

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***Abstract*—In air-to-ground surveillance with spaceborne or airborne radar systems, conventional along-track interferometry (ATI) directly computed from dual-channel synthetic aperture radar (SAR) (or range-Doppler) images may become unreliable for moving targets in the endo-clutter region due to the low signal-to-clutter-plus-noise ratio (SCNR), thereby resulting in degraded performance in the ATI-based detection. To address this issue, this article investigates an match filtered ATI (MF-ATI) and a pseudo-signal aided match filtered ATI (PSMF-ATI) for array radar ground moving target indication (GMTI) applications. For a N-channel radar system, the MF-ATI constructs an interferogram between two adaptively clutter-suppressed images derived from the first $N-1$ antenna channels (1 to $N-1$) and the last $N-1$ antenna channels (2 to N), respectively. The enhanced SCNR improves estimation accuracy of the interferometric phase of moving targets, which is linearly related to their radial velocity. To suppress noise contamination, a pseudo signal with power higher than the noise floor yet lower than that of the target is introduced to stabilize the interferometric phase around zero radians in the absence of moving targets, which is defined as PSMF-ATI. Contrast between the interferogram phase of moving targets and that of the detection background is significantly increased. Experimental results using real datasets verify the improvements in the interferometric phase estimation and detection for low-SCNR moving targets.**

***Index Terms*—Array radar, moving target detection, ATI, match filter, pseudo signal**

I. Introduction

In air-to-ground surveillance, synthetic aperture radar (SAR) ground moving target indication (GMTI) systems generally deploy an antenna array along the track direction to suppress the ground clutter for detecting weak moving targets in the endo-clutter region [1], [2]. In practice, ground clutter is typically heterogeneous and the clutter-suppression processing often suffers from performance degradation, leading to a high probability of false alarms (Pfa) in the magnitude-based detection methods [3]–[5].

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Increasing the magnitude detection threshold can reduce Pfa, while moving targets with low signal-to-clutter-plus-noise ratio (SCNR) values may have a low probability of target detection (Pd). The along-track interferometry (ATI) phase between dual-channel SAR images is theoretically proportional to the line-of-sight velocity of the backscatter, providing a crucial cue to distinguish moving targets from heterogeneous clutter and thereby improving SAR-GMTI performance in heterogeneous environments [6]–[10]. However, the ATI phases of low-SCNR targets are often affected by strong clutter signals, increasing the minimum discernible SCNRs required for successful target detection.

Recently, a moving target detection method that integrates the residual's magnitude from clutter suppression with the interferometric phase across two clutter-suppression residuals derived from the preceding and succeeding channel sets, respectively, has demonstrated improved performance [11]. However, since most residuals approximate noise signals in the absence of moving target, their interferometric phases tend to be randomly distributed within $[-\pi, \pi]$ radians (rad), which may amplify the combined test response in the absence of moving targets, thereby limiting the detection capacity.

To address this issue, this article investigates two novel ATI-phase tests: match filtered ATI (MF-ATI) and pseudo-signal aided match filtered ATI (PSMF-ATI), with array radar GMTI systems. For a N-channel radar system, the MF-ATI measures the interferogram between two clutter-suppression residuals derived from the first $N-1$ channels and the last $N-1$ channels, respectively, aiming to improve estimation accuracy of the interferometric phase for low-SCNR targets. Furthermore, the PSMF-ATI introduces a pseudo signal with power higher than the noise floor yet lower than that of the target in both residuals to stabilize the interferometric phase around zero radians in the absence of moving targets, thereby enhancing the contrast between the interferogram phase of moving targets and that of the detection background. The experimental results on real-world datasets are presented to

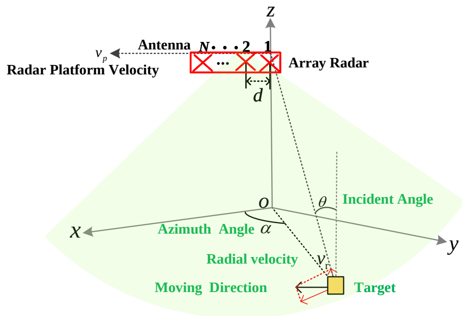

Fig. 1. Observation geometry illustrating relationship between the N-channel SAR system and ground moving targets in the side-looking GMTI mode.

demonstrate their potentials for GMTI.

Notations: The operators $*$, $(\cdot)^{\mathsf{T}}$, $(\cdot)^{\mathsf{H}}$, $|\cdot|$ and $E[\cdot]$ denote the complex conjugate, transpose and conjugate transpose, the modulus of a complex quantity, and the mathematical expectation, respectively. $\mathbf{I}$ is the identity matrix of appropriate dimension, and i is the imaginary unit ($\mathrm{i}^2 = -1$).

II. Signal Model of Array Radar System

Consider an airborne GMTI SAR equipped with a linear array of N antenna elements uniformly distributed along the platform's track direction with an inter-element spacing of d, operating in side-looking mode. The geometric relationship between the GMTI SAR and a moving target is illustrated in Fig.1. At a reference time, the origin o and the x axis correspond to the aircraft's ground-projected position and track (azimuth) direction, respectively. Without loss of generality, the z- and y-axes are defined as perpendicular to the ground and to the xoz-plane, respectively. Within a coherent processing interval (CPI), the aircraft is assumed to move with a constant velocity v_{p} along the x-axis. As shown in Fig.1, for a ground moving target, the azimuth and incident angles are expressed by α and θ, respectively, while, the radial velocity, obtained by projecting the target's motion onto the radar line-of-sight direction, is represented by v_{r}.

During ground-moving target detection, the radar transmits broadband electromagnetic waves through N channels over the CPI. Identical signal processing chains are employed to process the sampled data received by each antenna element. Under range-Doppler processing or SAR imaging, a slowly moving ground target is assumed to remain matched-filtered within the radar signal bandwidth and coherently integrated in the CPI, resulting in N range-Doppler maps. In each range-Doppler image, the along-track and cross-track directions correspond to azimuth and slant range respectively. Subsequently, the spatial alignment of the ground scatters across the N SAR images is performed. The constant amplitude and phase offsets among the SAR images, caused by fixed channel response errors of the array elements, and the varying phase offsets of the array elements, introduced by the platform's motions, are corrected via registration and calibration procedures.

Let $\mathbf{z}(k) = [z_1(k), z_2(k), \cdots, z_N(k)]^{\mathsf{T}}$ represent an $N \times 1$ data vector snapshot from the processed N SAR images, for a pixel $k \in \{1, 2, \cdots, K\}$. Here, $z_j(k)$ denotes the complex signal in the j-th SAR image obtained from the j-th channel, $j \in \{1, 2, \cdots, N\}$. Accordingly, the binary hypothesis test for detecting targets is defined as

$$\begin{aligned} \mathrm{H}_0 &: \mathbf{z}(k) = \mathbf{c}(k) + \mathbf{n}(k), \\ \mathrm{H}_1 &: \mathbf{z}(k) = \mathbf{s}(k) + \mathbf{c}(k) + \mathbf{n}(k), \end{aligned} \tag{1}$$

where H_0 is the target-absent hypothesis, and H_1 is the target-present hypothesis; $\mathbf{c}(k)$ and $\mathbf{s}(k)$ are the clutter vector and the signal vector of a deterministic moving target, respectively; the noise vector $\mathbf{n}(k)$ follows a zero-mean complex Gaussian distribution, denoted as $\mathbf{n}(k) \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{n}}^2 \mathbf{I})$ with σ_{n}^2 being the noise power.

Under the hypothesis H_1, the amplitudes of the focused target signal across all channels are assumed to be identical, such that $\mathbf{s}(k)$ can be expressed as

$$\mathbf{s}(k) = x_{\mathrm{t}}(k)\mathbf{a}_{\mathrm{t}}(k). \tag{2}$$

where $x_{\mathrm{t}}(k)$ denotes the complex target amplitude, and $\mathbf{a}_{\mathrm{t}}(k)$ denotes the target spatial steering vector. For a ground moving target with radial velocity $v_{\mathrm{r}}(k)$, as shown in Fig.1, the target Doppler $f_{\mathrm{t}}(k) = 2v_{\mathrm{r}}(k)/\lambda$ induces a phase shift of $2\pi f_{\mathrm{t}}(k)\frac{d/2}{v_{\mathrm{p}}}$ as the array traverses the baseline spacing d, where λ is the radar wavelength. Then, $\mathbf{a}_{\mathrm{t}}(k)$ is given by

$$\begin{aligned} &\mathbf{a}_{\mathrm{t}}(k) \\ &= \left[1, \exp\left(\mathrm{i}2\pi \frac{f_{\mathrm{t}}(k)d}{2v_{\mathrm{p}}}\right), \cdots, \exp\left(\mathrm{i}2\pi \frac{f_{\mathrm{t}}(k)(N-1)d}{2v_{\mathrm{p}}}\right)\right]^{\mathsf{T}}. \end{aligned} \tag{3}$$

The clutter signal vector follows the form

$$\mathbf{c}(k) = x_{\mathrm{c}}(k)\mathbf{a}_{\mathrm{c}}(k), \tag{4}$$

where $x_{\mathrm{c}}(k)$ denotes the complex amplitude of ground backscatter, and the statistics are often non-uniform due to the composite scattering features in realistic conditions, while $\mathbf{a}_{\mathrm{c}}(k)$ is the spatial steering vector associated with clutter, and $\mathbf{a}_{\mathrm{c}}(k) \approx [1, \cdots, 1]^{\mathsf{T}}$ for ground clutter.

The classical complex multilook interferometry is defined based on dual-channel SAR images as [7]

$$I(k) = \frac{\frac{1}{L}\sum_{l=1}^{l=L} z_1(l) z_2^*(l)}{\sqrt{\mathrm{E}\left[z_1 z_1^*\right] \mathrm{E}\left[z_2 z_2^*\right]}}, \tag{5}$$

where L denotes the number of looks, and $z_1(l)$ and $z_2(l)$ are the l-th look data of the pixel k in the first and the second channels, respectively, with independent and identical statistics over L looks. $|I(k)|$ and $\phi(k) = \arg\left[I(k)\right]$ are defined as the normalized interferometric magnitude and phase, respectively, with $\phi(k)$ confined within $[-\pi, \pi]$.

III. Two Reformed ATI Techniques

A. MF-ATI

The SAR images from the first $N-1$ and the latter $N-1$ channels are first used to construct two data vectors for a given pixel k as follows

$$\mathbf{z}_1(k) = \left[z_1(k), z_2(k), \cdots, z_{N-1}(k)\right]^{\mathsf{T}}, \tag{6a}$$

$$\mathbf{z}_2(k) = \left[z_2(k), z_3(k), \cdots, z_N(k)\right]^{\mathsf{T}}, \tag{6b}$$

which differ in phase due to the time delay $d/(2v_\mathrm{p})$ as the platform moves across the channel spacing d. According to the signal model in (1), the data vectors in (6) can be rewritten as $\mathbf{z}_1(k) = \mathbf{c}_1(k) + \mathbf{n}_1(k)$ and $\mathbf{z}_2(k) = \mathbf{c}_2(k) + \mathbf{n}_2(k)$ under H_0, while they become $\mathbf{z}_1(k) = \mathbf{s}_1(k) + \mathbf{c}_1(k) + \mathbf{n}_1(k)$ and $\mathbf{z}_2(k) = \mathbf{s}_2(k)+\mathbf{c}_2(k)+\mathbf{n}_2(k)$ under H_1. Here, $\mathbf{s}_1$, $\mathbf{c}_1$, and $\mathbf{n}_1$ denote the target, clutter and noise signals in $\mathbf{z}_1$, respectively, while $\mathbf{s}_2$, $\mathbf{c}_2$, and $\mathbf{n}_2$ correspond to those in $\mathbf{z}_2$. In the SAR images, the phase difference between the two signal vectors during the time delay $d/(2v_\mathrm{p})$ is proportional to the Doppler frequency (or radial velocity) of the scatters:

$$\mathbf{c}_2(k) = \mathbf{c}_1(k), \quad \mathbf{s}_2(k) = \mathbf{s}_1(k) \exp\left(-\mathrm{i}2\pi f_\mathrm{t}(k)\frac{d}{2v_\mathrm{p}}\right). \quad (7)$$

Note that the deterministic phase terms in (7) do not affect the respective spatial steering vectors, thus allowing the same filtering processor $\mathbf{u}_1(k)$ to be applied to both data vectors for clutter suppression

$$y_1(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{z}_1(k), \qquad y_2(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{z}_2(k), \quad (8)$$

where $\mathbf{u}_1(k)$ denotes the normalized weight vector for adaptive clutter suppression satisfying $\mathbf{u}_1^\mathsf{H}\mathbf{u}_1 = 1$ [11]. Let $y_{\mathrm{c}1}(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{c}_1(k)$ and $y_{\mathrm{s}1}(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{s}_1(k)$ denote the clutter and target residuals, respectively; $y_{\mathrm{n}1}(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{n}_1(k)$ and $y_{\mathrm{n}2}(k) = \mathbf{u}_1^\mathsf{H}(k)\mathbf{n}_2(k)$ denote noise residuals in $\mathbf{z}_1(k)$ and $\mathbf{z}_2(k)$, respectively. As $\mathbf{u}_1^\mathsf{H}\mathbf{u}_1 = 1$, $\mathbf{n}_1(k) \sim \mathcal{CN}(\mathbf{0}, \sigma_\mathrm{n}^2\mathbf{I})$, and $\mathbf{n}_2(k) \sim \mathcal{CN}(\mathbf{0}, \sigma_\mathrm{n}^2\mathbf{I})$, two residual images have an equal noise floor σ_n^2.

By applying the ATI processing between residual images y_1 and y_2, the MF-ATI phase is given as

$$\phi_\mathrm{MF}(k) = \arg\left[\sum_{l=1}^{l=L} y_1(l)y_2^*(l)\right], \quad (9)$$

where L looks are often selected in the vicinity of pixel k in the processing, and $y_1(l)$ denotes the l-th look of pixel k in the first residual image, while $y_2(l)$ corresponds to the l-th look of pixel k in the second residual image.

Under H_1, since most clutter can be effectively suppressed, leading to $\mathrm{E}\left[|y_{\mathrm{c}1}|\right] \approx 0$, the target residuals, whose magnitudes exceed those of the noise, play a dominant role in the operation, leading to

$$\phi_\mathrm{MF}(k) \approx 2\pi f_\mathrm{t}(k)\frac{d}{2v_\mathrm{p}} = 2\pi\frac{dv_\mathrm{r}(k)}{\lambda v_\mathrm{p}}. \quad (10)$$

Note that the phase wrapping across a period of 2π will occur in $\phi(k)$ if the target's radial velocity $v_\mathrm{r}(k)$ exceeds the unambiguous velocity range $[-\lambda v_\mathrm{p}/(2d), \lambda v_\mathrm{p}/(2d)]$.

Under H_0, clutter residuals may dominate the noise signal due to the presence of heterogeneous clutter, resulting in $\mathrm{E}\left[|y_{\mathrm{c}1}|^2\right] \gg \sigma_\mathrm{n}^2$. Accordingly, the interferometric phase remains close to 0 radians, allowing moving targets to be distinguished from the clutter. However, in the other case of H_0, clutter suppression may become highly effective for homogeneous backgrounds, such that the noise signals dominate residuals, where $\mathrm{E}\left[|y_{\mathrm{c}1}|^2\right] \ll \sigma_\mathrm{n}^2$. The interferometric phase values associated with the noise tends to be randomly distributed over the range $[-\pi, \pi]$, competing with true moving targets in phase-based detection.

B. PSMF-ATI

To suppress the noise contamination in MF-ATI, a pseudo signal following the zero-mean and $k_\mathrm{c}\tilde{\sigma}_\mathrm{n}^2$-variance Gaussian distribution, such as $\tilde{b}(k) \sim \mathcal{CN}(0, k_\mathrm{c}\tilde{\sigma}_\mathrm{n}^2)$, is introduced into both residuals

$$\tilde{y}_1(k) = y_1(k) + \tilde{b}(k), \quad \tilde{y}_2(k) = y_2(k) + \tilde{b}(k), \quad (11)$$

where k_c is a constant in the range $(1, 10]$ to satisfy $\sigma_\mathrm{n}^2 < \mathrm{E}[|\tilde{b}|^2] < \sigma_\mathrm{s}^2$, and σ_s^2 denotes the target residual power, typically greater than ten times of the noise.

Next, applying the complex L-look interferogram over $\tilde{y}_1(k)$ and $\tilde{y}_2(k)$ yields the PSMF-ATI phase

$$\phi_\mathrm{PSMF}(k) = \arg\left[\sum_{l=1}^{l=L} \tilde{y}_1(l)\tilde{y}_2^*(l)\right]. \quad (12)$$

Using clutter, target and noise components in the two residuals, (12) can be reformulated as (13). Under H_0, the clutter-dominated (or pseudo-signal-dominated) case that satisfies $\mathrm{E}\left[|y_{\mathrm{c}1}|^2\right] \gg k_\mathrm{c}\sigma_\mathrm{n}^2$ (or $\mathrm{E}\left[|y_{\mathrm{c}1}|^2\right] \ll k_\mathrm{c}\sigma_\mathrm{n}^2$) is used to compute the interferometric phase, leading to (13b), where the interferometric phase $\phi_\mathrm{PSMF}(k)$ remains close to 0 radians under all conditions. Under H_1, since the target residuals still play a dominant role in the PSMF-ATI, the computed interferometric phase $\phi_\mathrm{PSMF}(k)$ approximately varies with the target's radial velocity. When $k_\mathrm{c} = 0$, PSMF-ATI could degenerate into MF-ATI.

IV. Experimental Results

To evaluate the MF-ATI and PSMF-ATI tests, the real data, collected by an airborne four-channel X-band SAR with an inter-channel spacing of 0.25 m were used in the experiments. The raw SAR data containing 512 pulses at a pulse repetition frequency of 800 Hz were processed for imaging, channel coregistration, and balancing, leading to four SAR images of the same scene with a spatial resolution of 3 m $\times$ 3 m (slant range $\times$ azimuth). Since the true positions of real-world ground moving targets are unknown, eight synthetic targets were simulated based on the signal models in (2) and (3), and then embedded into the real clutter background. Each simulated target occupies 9 pixels around its center positions in the SAR image and has varying radial velocities and SNR values: (15 dB, 1.5 m/s), (15 dB, 2.0 m/s), (15 dB, 2.5 m/s), (15 dB, 3.0 m/s), (25 dB, 1.5 m/s), (25 dB, 2.0 m/s), (25 dB, 2.5 m/s), and (25 dB, 3.0 m/s), respectively.

Comparisons of the conventional ATI result, MF-ATI result, and PSMF-ATI result are shown in Fig. 2. In Fig. 2(a), the interferometric magnitude derived from the SAR images contains extensive strong clutter from environments; however, in Fig. 2(c) and Fig. 2(e), most clutter has been effectively suppressed, enabling the simulated moving targets marked by red circles to become distinguishable from the background. In Fig. 2(b), the interferometric phases of stationary ground clutter are mainly distributed around 0 rad, whereas those of

$$
\mathrm{H}_0: \; \phi_{\mathrm{PSMF}}(k) = \arg\left[\sum_{l=1}^{l=L}\left(y_{\mathrm{c1}}(l) + y_{\mathrm{n1}}(l) + \tilde{b}(l)\right)\left(y_{\mathrm{c1}}(l) + y_{\mathrm{n2}}(l) + \tilde{b}(l)\right)^*\right] \tag{13a}
$$

$$
\approx \begin{cases} \arg\left[\sum_{l=1}^{l=L} y_{\mathrm{c1}}(l) y_{\mathrm{c1}}^*(l)\right] \approx 0, & \text{if} \quad \mathrm{E}\left[|y_{\mathrm{c1}}|^2\right] \gg k_{\mathrm{c}}\sigma_{\mathrm{n}}^2 \\ \arg\left[\sum_{l=1}^{l=L} \tilde{b}(l)\tilde{b}^*(l)\right] = 0, & \text{if} \quad \mathrm{E}\left[|y_{\mathrm{c1}}|^2\right] \ll k_{\mathrm{c}}\sigma_{\mathrm{n}}^2 \\ \arg\left[\sum_{l=1}^{l=L}\left(y_{\mathrm{c1}}(l) y_{\mathrm{c1}}^*(l) + \tilde{b}(l)\tilde{b}^*(l)\right)\right] \approx 0, & \text{otherwise.} \end{cases} \tag{13b}
$$

$$
\mathrm{H}_1: \; \phi_{\mathrm{PSMF}}(k) = \arg\left[\sum_{l=1}^{l=L}\left(y_{\mathrm{s1}}(l) + y_{\mathrm{c1}}(l) + y_{\mathrm{n1}}(l) + \tilde{b}(l)\right)\left(y_{\mathrm{s1}}(l)\exp\left(-\mathrm{i}2\pi f_{\mathrm{t}}(l)\frac{d}{2v_{\mathrm{p}}}\right) + y_{\mathrm{c1}}(l) + y_{\mathrm{n2}}(l) + \tilde{b}(l)\right)^*\right] \tag{13c}
$$

$$
\approx \arg\left[\sum_{l=1}^{l=L} y_{\mathrm{s1}}(l) y_{\mathrm{s1}}^*(l)\exp\left(\mathrm{i}2\pi f_{\mathrm{t}}(l)\frac{d}{2v_{\mathrm{p}}}\right)\right] \approx 2\pi f_{\mathrm{t}}(k)\frac{d}{2v_{\mathrm{p}}} = 2\pi\frac{v_{\mathrm{r}}(k)d}{\lambda v_{\mathrm{p}}}. \tag{13d}
$$

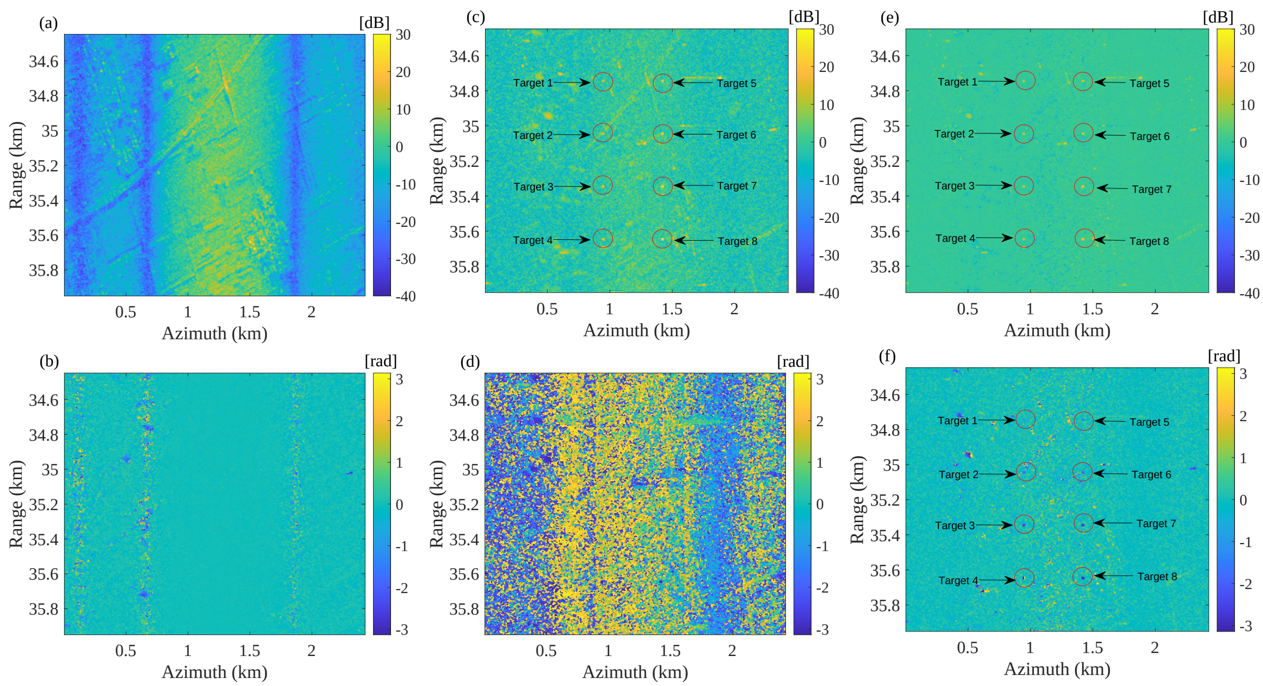

Fig. 2. Comparison of the complex interferogram: (a), (b) five-look interferometric magnitude and phase between SAR images from channels 1 and 2; (c), (d) MF-ATI magnitude and phase; (e), (f) PSMF-ATI magnitude and phase with $k_{\mathrm{c}} = 6$ dB.

the simulated targets are difficult to distinguish due to their low SCNR ranging from -30 dB to -20 dB. In Fig. 2(d), most residuals behave as noise-like signals, and the MF-ATI phases are randomly distributed in the range $[-\pi, \pi]$ rad, rendering the simulated targets undetectable from the background. In Fig. 2(f), the PSMF-ATI phases of the clutter residuals are generally distributed around 0 rad, and those of the simulated targets stand out clearly from the background.

V. Conclusion

In this article, the proposed MF-ATI measures the interferometric phase between clutter-suppressed images from the preceding and succeeding channels for array radar ground moving target indication (GMTI) applications. Based on the MF-ATI, PSMF-ATI further introduces a pseudo signal with power higher than the noise floor yet lower than that of the target into two residuals to stabilize the interferogram phase around zero radians in the absence of moving targets. The enhanced signal-to-clutter-plus-noise ratio (SCNR) improves estimation accuracy of the interferogram phase of moving targets, which is linearly related to their radial velocity. Experimental results demonstrate the significant features of the two novel ATI techniques. Compared with MF-ATI, PSMF-ATI enables moving targets to be clearly detectable from the background. Our future work would focus on the study of MF-ATI and PSMF-ATI for radar GMTI applications.

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